Project Title: Spider Quadruped Robot using ESP32

Short Description:

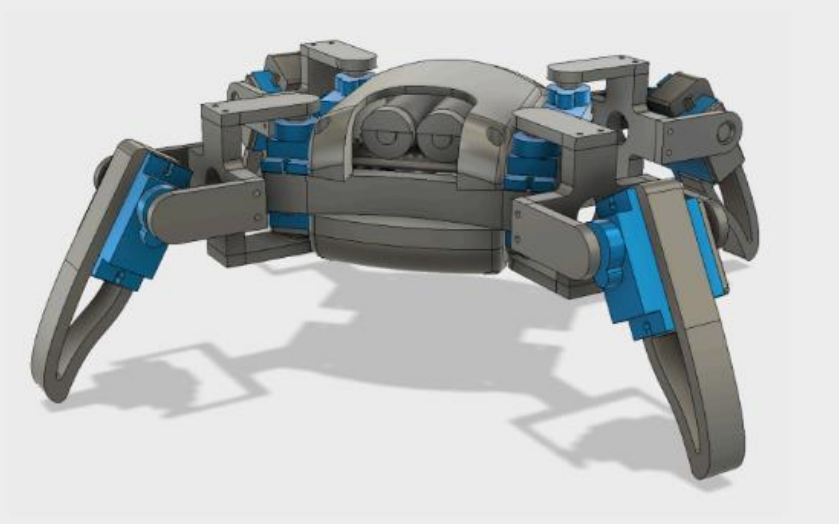
This project is a **4-legged spider robot** powered by an **ESP32 Dev Board** to control 8 SG90 servos (2 per leg). The bot is capable of walking, rotating, and performing smooth movements via programmed servo coordination. Designed using 3D-printed parts, it is wirelessly controllable and energy-efficient, running on Li-ion batteries regulated with a **DC-DC Step-Down Converter (MP1584).**

Component List:

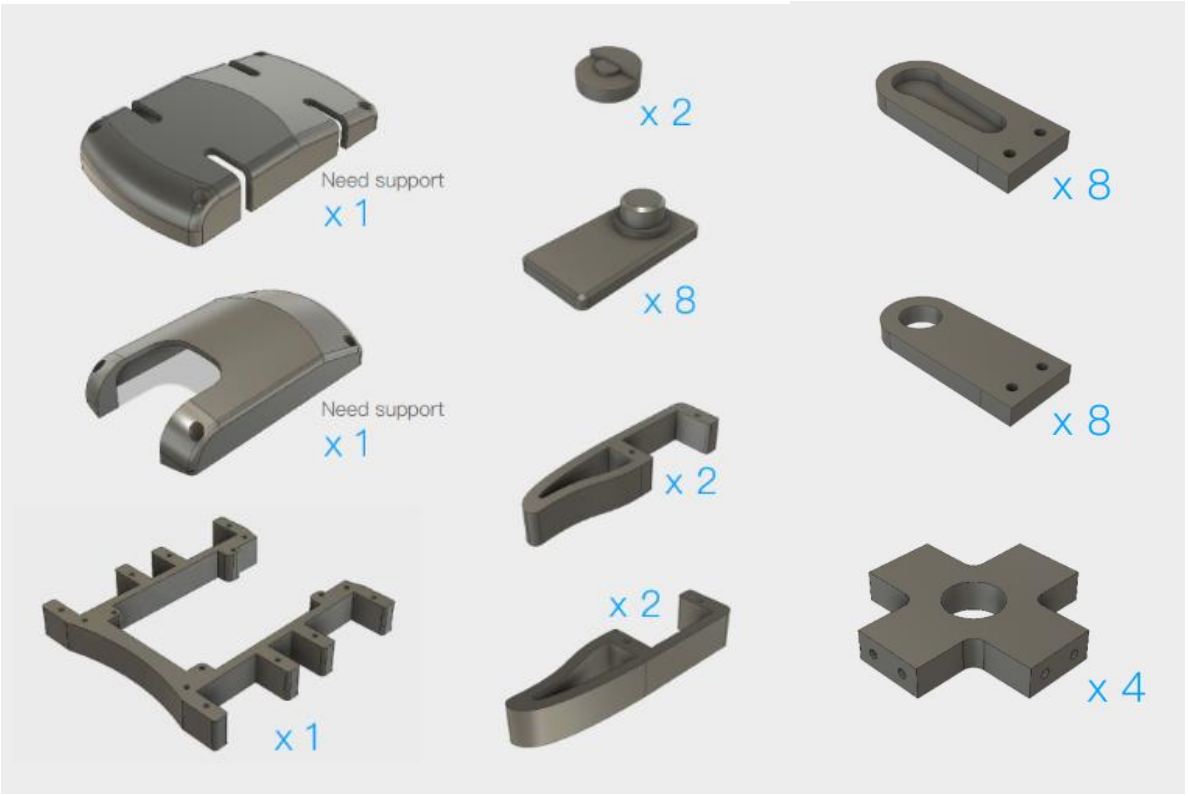
Component	Quantity	Description
ESP32 Dev Board	1	Main microcontroller with WiFi + Bluetooth
TowerPro MG90S Servo Motors	8	Controls hip and knee joints (2 per leg × 4 legs)
3D Printed Parts	29	Robot frame, joints, servo mounts, leg parts
M1.7 x 8mm Screws	~60	For mounting servos and joints
MP1584 Step-Down Converter	1	Converts 7.4V (from 2× Li-ion) down to stable 5V output
Battery Pack (2× Li-ion 3.7V 18650)	1	Provides 7.4V input to MP1584 converter

Download STL Files:

SLT Link - https://www.thingiverse.com/thing:2311678



3D Print Parts:



Circuit Diagram:

